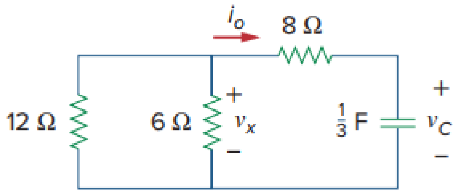
Chapter 7, Problem 1PP

Problem

Refer to the circuit in Fig. 7.7. Let $vC(0)$ = 60 V. Determine vC, vx, and io for $t \geq 0$.

Fig. 7.7:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

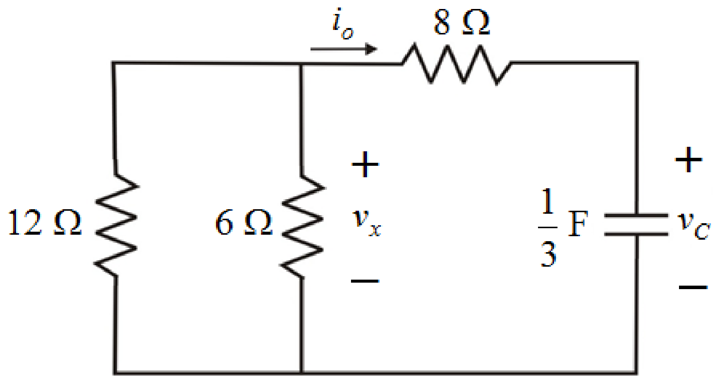


Figure 1

Step 2 of 6

Resistors $12\ \Omega$ and $6\ \Omega$ shown in Figure 1 are connected in parallel. Calculate an equivalent resistance of parallel resistors.

$$R = \frac{(12)(6)}{12+6}$$
$$= \frac{72}{18}$$
$$= 4\ \Omega$$

The parallel equivalent resistance is in series with $8\ \Omega$ resistor. Calculate the total equivalent resistance.

$$R_{eq} = 4+8$$
$$= 12\ \Omega$$

Step 3 of 6

Draw the modified circuit diagram after converting parallel series resistors into equivalent.

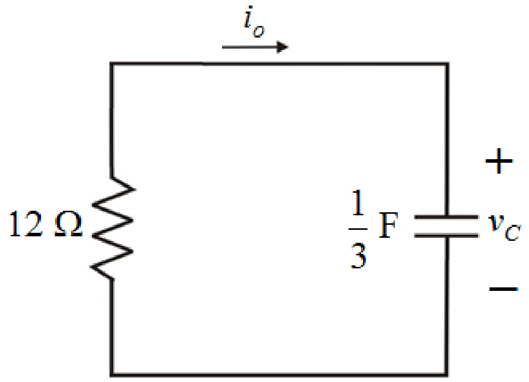


Figure 2

Step 4 of 6

Calculate the time constant.

$$\begin{aligned}\tau &= R_{eq}C\\ &= (12)\left(\frac{1}{3}\right)\\ &= 4\text{ s}\end{aligned}$$

Calculate the voltage across the capacitor for $t > 0$.

$$\begin{aligned}v_C(t) &= v_C(0)e^{-\frac{t}{\tau}}\\ &= (60)e^{-\frac{t}{4}}\\ &= 60e^{-0.25t}\text{ V}\end{aligned}$$

Therefore, the voltage across capacitor, $v_C(t)$ is $\boxed{60e^{-0.25t}\text{ V}}$.

Step 5 of 6

Apply Kirchhoff's current law at node v_x to the Figure 1.

$$\frac{v_x}{12} + \frac{v_x}{6} + \frac{v_x - v_C}{8} = 0$$

$$\left(\frac{1}{12} + \frac{1}{6} + \frac{1}{8}\right)v_x = \frac{v_C}{8}$$

$$\left(\frac{2+4+3}{24}\right)v_x = \frac{60e^{-0.25t}}{8}$$

$$\begin{aligned}v_x &= \frac{(24)\left(60e^{-0.25t}\right)}{(8)(9)}\\ &= 20e^{-0.25t}\text{ V}\end{aligned}$$

Therefore, the voltage across resistor, v_x is $\boxed{20e^{-0.25t}\text{ V}}$.

Step 6 of 6

Calculate the current through $8\,\Omega$ resistor.

$$\begin{aligned} i_o(t) &= \frac{v_x(t) - v_c(t)}{8} \\ &= \frac{20e^{-0.25t} - 60e^{-0.25t}}{8} \\ &= -\frac{40}{8}e^{-0.25t} \\ &= -5e^{-0.25t}\ \text{A} \end{aligned}$$

Therefore, the current through $8\,\Omega$ resistor, i_o is $\boxed{-5e^{-0.25t}\ \text{A}}$.